

“ADAPTIVE AIR QUALITY FORECASTING USING ENSEMBLE LEARNING AND CTGAN WITH EXPLAINABLE AI”

Major Project Report

*Submitted in Partial Fulfillment of the
Requirements for the Degree of*

BACHELOR OF TECHNOLOGY

IN

**INFORMATION AND COMMUNICATION
TECHNOLOGY (ICT)**

By
Keya Padia
22BIT048

Under the Guidance of
Ms. Aarti Solanki



**Department of Information and Communication Technology,
School of Technology, Pandit Deendayal Energy University,
Gandhinagar 382 426**

May 2026

Certificate of Originality of Work

I hereby declare that the B.Tech. Project entitled “**Adaptive Air Quality Forecasting Using Ensemble Learning And CTGAN With Explainable AI**” submitted by me for the partial fulfillment of the degree of **Bachelor of Technology** to the **Dept. of Information and Communication Technology** at the **School of Technology, Pandit Deendayal Energy University**, Gandhinagar, is the original record of the project work carried out by me under the supervision of **Ms. Aarti Solanki**.

I also declare that this written submission adheres to university guidelines for its originality, and proper citations and references have been included wherever required.

I also declare that I have maintained high academic honesty and integrity and have not falsified any data in my submission.

I also understand that violation of any guidelines in this regard will attract disciplinary action by the institute.

Name of the Student: Keya Padia

Roll Number of the Student: 22BIT048

Signature of the Student:

Name of the Supervisor: Ms. Aarti Solanki

Designation of the Supervisor: Assistant Professor

Signature of the Supervisor:

Place: Gandhinagar

Date: 20/05/2026

Certificate from the Project Supervisor

This is to certify that the Major Project Report entitled “**Adaptive Air Quality Forecasting Using Ensemble Learning and CTGAN With Explainable AI**” submitted by **Ms. Keya Padia**, Roll No. **22BIT048** towards the partial fulfilment of the requirements for the award of degree in **Bachelor of Technology** in the field of **Information and Communication Technology Engineering** from the **School of Technology, Pandit Deendayal Energy University, Gandhinagar** is the record of work carried out by her under my supervision and guidance. The work submitted by the student in my opinion reached a level required for being accepted for examination. The results embodied in this major project work to the best of our knowledge have not been submitted to any other University or Institution for the award of any degree or diploma.

Ms Aarti Solanki
Name and Sign of the Supervisor

Dr. Paawan Sharma
Name and Sign of the HoD

Prof. Dr. Anirbid Sircar
Name and Sign of the Director

Place: Gandhinagar

Date: 20/05/2026

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Keya Padia

Abstract

At present, environmental sensor networks are used to monitor air quality. However, creating accurate models for this process remains difficult. In many cases, datasets are imbalanced and contain a massive number of missing values. In addition to this, many researchers have viewed predictive models as black boxes, and this makes it impossible to explain how predictions were made.

CTGAN to handle minority sampling and a confidence score mechanism to manage missing values will be used to create a new framework for predicting AQI values. This will be further combined with ensemble learning and SHAP analysis. To enhance the robustness and accuracy of the model, some additional steps will be used, such as feature engineering, time interpolation and real-time dashboards.

The study also focuses on a special study of the top five polluted cities of the country – Gurugram, Delhi, Patna, Lucknow and Kolkata, to further validate the effectiveness of the framework in extreme conditions. The research examines the localized nature of air pollutants and the effect of geographical factors in different regions on the accuracy of the model by using cross-city transferability matrices. This complete assessment shows that a global ensemble approach can give a good baseline but it is important to refine it on a local scale to deal with the unpredictable high peaks of AQI typical of these high pollution urban areas.

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NOMENCLATURE

C	Hourly concentration of the pollutant
BP	Concentration breakpoint
I	AQI value corresponding to BP
AQI	Calculated Air Quality Index
Y	Final predicted AQI value
X	Input feature vector

Greek

φ	Shapley value
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Subscripts

p	Pollutant type
hi	Upper bound
lo	Lower bound
XGBoost	XGBoost model component
LightGBM	LightGBM model component

Abbreviations

AQI	Air Quality Index
CTGAN	Conditional Tabular Generative Adversarial Network
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
SHAP	Shapley Additive Explanations
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
PM2.5	Particulate Matter with diameter less than 2.5 micrometers
PM10	Particulate Matter with diameter less than 10 micrometers

Chapter 1

Introduction

1.1 Background

In the current urban environment, air pollution continues to have a serious impact on the public's health. It is also leading to environmental issues, affecting many rapidly urbanizing cities around the world, and causing severe health effects including respiratory diseases, cardiovascular issues, and diminished life quality. These environmental changes are widely monitored using sensor networks to gather continuous data about different atmospheric pollutants.

Governments and environmental agencies use the Air Quality Index (AQI) to communicate this intricate scientific information to the public in a simple and understandable manner. This index is a standardized tool for presenting the concentration data in the form of simple color-coded indicator ratings that include Good, Moderate, Poor and Hazardous. This classification enables instant risk evaluation, and enables the public to see at a glance what the air quality is like.

The major criteria pollutants of the AQI - those with the greatest health impacts - are Particulate Matter (PM_{2.5} and PM₁₀), Nitrogen Dioxide (NO₂), Sulfur Dioxide (SO₂), Carbon Monoxide (CO), and Ozone (O₃). The AQI is assigned to each pollutant based on a set of standardized AQI breakpoint ranges established by international regulatory agencies, such as U.S. Environmental Protection Agency (EPA). Early warning and forecasting of these indices is critical for health action and urban planning.

1.2 Motivation

The main technical reason for choosing this project is that air pollution sensors often malfunction or stop working for long periods. This results in data with many gaps and inconsistencies that are very hard to handle. Standard ways of fixing missing data often fail to catch real changes in the air, making the final results unreliable. Also, there aren't many research papers that study both the global air quality situation and the very unique, high-pollution patterns found in Indian cities. I wanted to build a framework that can handle these real-world data problems and still give reliable predictions even when the sensor data is poor.

Beyond just chasing high accuracy scores, I wanted to create a system that people can actually trust and understand. Many current models act like "black boxes" where we don't know how they reach a conclusion, which makes them hard to use in real life. My focus is on making a reliable system that can identify rare but dangerous pollution events that simpler models usually miss. Since so many cities are currently struggling with toxic air, I felt a strong need to contribute to environmental betterment. By providing a tool that is both accurate and explainable, this project aims to give better information to the public and help reduce the health problems caused by poor air quality.

1.3 Objectives

- **Robust & Accurate Forecasting:** To construct a stacking ensemble model (XGBoost, LightGBM) strengthened by synthetic data generated by CTGAN with the aim of increasing stability of prediction and addressing data scarcity under rare and extreme hazard scenarios.
- **Adaptive Estimation:** Design a confidence-aware AQI mechanism that can ensure the operation reliability when sensor failure occurs, and determine the health risk according to the AQI based on the available readings without using unreliable interpolation among sensors.
- **Explainable & Actionable Analytics:** Combine the explainability of SHAP with a real-time Power BI dashboard to make complex chemical triggers more transparent and color-coded, enabling actionable insights for informed decision making.

1.4 Problem Statement

The first big problem is that environmental sensors are unreliable. In the real world, these devices are prone to being tripped, malfunctioning or running out of battery for extended periods of time. This results in large data gaps and standard models are unable to make a continuous forecast. Without the data, the entire system collapses, and most existing systems don't have an effective means of continuing to accurately function during the "sensor blackouts".

The second issue is how data that has been lost is dealt with. The majority of researchers apply the basic technique of filling in the blanks with the mean (average) or median (middle) for the day. This can be done easily, but it is extremely hazardous because it introduces "fake" patterns. Smooths and obscures real pollution peaks. This can render the model to think the air is safe when it is not, causing a false sense of security.

The third problem is the high degree of imbalance of air quality data sets. Most cities have many "Good" or "Moderate" days and comparatively few "Hazardous" or "Severe" days. Machine learning models learn from examples, and they become extremely proficient at predicting normal days, but not the rare hazardous conditions. This is a major issue because the primary purpose of an AQI model is to alert the public to situations when air quality becomes unhealthy and not when it is normal.

Lastly, there's the "black box" issue with contemporary AI. There are more advanced models that are able to be very accurate but do not give an explanation of why a prediction was made. When a model indicates that AQI is going to jump high, one will need to understand whether it is due to an increase in Dust (PM10) or industrial gases (SO₂) to take the appropriate measures. This transparency is essential for people to have faith in the system. Also, models developed in one city may not be applicable in another, and display geographic fragility.

1.5 Approach

To address the challenges identified above, this project takes a step-wise strategy, starting with global testing and then gradually proceeding in a more local analysis. It consists of three major stages to guarantee the accuracy and reliability of the model.

The first part is a historical dataset of pollution across the world (OpenAQ 2016-2022) [1]. First, a "baseline" version of the model is built to observe the performance of standard methods. In this step, only basic data cleaning, dealing with outliers and calculation of AQI values are conducted. Then we train a simple XGBoost model and obtain an initial accuracy score. This is used as a benchmark against which to compare a variety of advanced forecasting models.

In the second phase, the proposed “Proposed Framework” is used to optimize the system. This involves more complex actions such as Time Interpolation of the sensors or Feature Engineering of the information to improve the input variables for the model. We also employ the CTGAN method for creating simulated pollution data for rare pollution events and the Ensemble Learning method (XGBoost + LightGBM) to achieve greater stability. To visualize the results, we present the results to make the results clear and explain the decisions of the model and connect it to a real-time visualization panel of Power BI dashboard.

In this last stage, the study is done on an Indian context based on a recent Kaggle dataset (2023-2025) [2]. The baseline and optimized steps are repeated for this data to check the generalization of the framework in India. To put this to the test, the top five most polluted cities in India are compared in detail as a final stress test. This comprises a spatial analysis as well as a model transferability matrix that determines whether a model developed in one city can predict the air quality of another, thus determining the robustness of the system or if it is city-specific.

1.6 Scope of the Project

The aim of this project is to develop and evaluate a robust global to local air quality forecasting system. This study focuses only on the Air Quality Index (AQI) and the major pollutants which affect AQI which are PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃. These are the basic criteria pollutants that are considered by consensus in the global context, such as EPA, to be most important to public health, and therefore the subject of the research.

The project's geographical and temporal scope is limited to tabular data. First, the model is trained and tuned using OpenAQ's large-scale historical data from 2016 to 2022 [1], to perform at a global scale. Second, it gets down to the Indian subcontinent on the basis of recent data 2023-2025 [2]. One of the key elements of the project is a deep-dive analysis of five cities in India: Gurugram, Delhi, Patna, Lucknow, and Kolkata. It enables the study to assess the model's performance during high pollution events and regional differences. Other form of datasets like satellite images, metrological data is out of the scope for this project.

The project should technically only concern itself with machine learning methods, namely ensemble methods and generative models for data balancing. It does not contain deep learning architectures such as LSTMs or CNNs, instead it's about building an accurate and interpretable system using SHAP analysis. All of this results in the production of a visual dashboard in Power BI to share these findings in a manner accessible to the general public.

1.7 Organization for the Rest of the Report

The remainder of this report is structured in several chapters to give an in-depth analysis of the research process; from the initial review of existing work to the final results.

Chapter 2: Literature Review & Problem Identification dives into the current research in air quality forecasting. It highlights other research literature and the technical challenges of this project – for example, the "black-box" problem of models and the lack of ability to deal with missing data.

Chapter 3: Proposed Methodology and Implementation explain the core technical framework in detail. Here we discuss the data preprocessing steps, the use of CTGAN to generate synthetic data, and how the ensemble learning approach works. It also provides a description of the environment used for the simulations and the datasets employed to create the models and test them.

Chapter 4: Results show the results that have occurred in practice with the framework proposed. It provides an optimized model, which when compared against the baseline, is validated with global and Indian data sets. This chapter also presents a 'deep dive' analysis of the top 5 most polluted cities in India, and the results of the model transferability matrix.

Chapter 5: Conclusions and Future Work provide a summary of the project's key findings and discusses its practical impact. It also recommends possible improvements and/or extensions to the framework in the future. The report ends with an Appendix, where other data is provided, and a Reference section, where all the literature cited is provided.

Chapter 2

Literature Review and Problem Identification

2.1 Literature Review

Research in air quality forecasting has evolved significantly, moving from basic statistical equations to complex AI-driven systems. This review examines how different researchers have handled the challenges of predicting the Air Quality Index (AQI) and where current technology stands.

Evolution from Statistical to Machine Learning Models: Early research relied heavily on traditional statistical methods like Linear Regression and Support Vector Machines [3]. Studies in the *Journal of Engineering Research* (2021) [4] used ARIMA to identify seasonal trends, while others compared it with Facebook's Prophet and LSTM [5], finding that while statistical models are easy to use, they lack the precision needed for fine particulate matter (PM2.5). This led to the adoption of machine learning. Researchers have successfully used XGBoost [6,7] and Random Forest [7,8] to handle the non-linear nature of pollution. Studies in *Environmental Science and Pollution Research* [9] further showed that hybrid models combining multiple algorithms consistently outperform single-model approaches when dealing with volatile urban data.

Handling Missing Data and Dataset Imbalance: A major hurdle in this field is "dirty data." Recent studies in *Symmetry* (2023) [10] and *PLOS ONE* (2024) [11] highlight that simple methods like mean-filling often introduce bias. To combat this, researchers have used "Median-KNN Regressors" [7] and "SMOTE-Tomek Links" [6,7] to balance datasets where "Hazardous" days are rare. A unique "cooperative" station system was even proposed in *Nature* (2024) [12], where stations "borrow" data from neighbors using a Gravity Model if their sensors fail. These advancements show a shift toward more sophisticated data cleaning that preserves the actual "spikes" in pollution rather than smoothing them out.

The Shift Toward Ensemble Methods: For large-scale data, many researchers are now moving toward "Ensemble Stacking." A 2025 study in *iScience* [13] demonstrated that stacking XGBoost and LightGBM provides the most stable predictions for Indian cities

like Hyderabad. Similar work in *Alexandria Engineering Journal* (2024) [10] suggests that while deep learning (LSTMs) is powerful, ensemble methods are often more practical for real-time city monitoring. This is especially relevant in India, where pollution patterns are highly regional.

Explainability and Transparent AI: The newest trend in air quality research is moving away from "black-box" models. A review in *Science of the Total Environment* (2025) [14] points out that for policy-making, we must understand *why* a model predicts a hazardous alert. Newer papers [15] have started integrating SHAP (Shapley Additive Explanations) to identify whether a spike is caused by industrial gases or traffic. This ensures that the AI is not just a math tool but an interpretable system that provides actionable insights for environmental betterment.

Multimodal integration of satellite imagery and deep learning architectures: The latest trend in air quality which places the whole emphasis away from the traditional tabular ground-based input to multimodal architectures that combine regional monitoring nodes with remote-sensing orbital imagery. Recent state-of-the-art frameworks published in *Scientific Reports* [16] uses advanced Convolutional Neural Network (CNNs) to extract high-dimensional feature map from visual satellite data, and then applies Bidirectional Long Short-Term Memory (BiLSTM) modules to extract fine-grained temporal patterns. Multi-modal networks employ deep attention mechanisms to dynamically observe ground sensor arrays together with satellite observations of Aerosol Optical Depth (AOD) from various platforms, such as Sentinel-5P, MODIS and Landsat-8 [17] to map complex spatial boundaries and to cover data gaps between the ground monitoring stations. These hybrid CNN-BiLSTM networks are in turn connected to Graph Neural Networks (GNNs), which can mathematically model physical distance and pollution transit paths between neighboring urban centers and deliver high accuracy, spatial continuity across entire regional ecosystems [16].

2.2 Problem Identification

Although much progress has been made, the literature also indicates that there are several critical gaps that prevent these models from being used for real-time, high-stakes applications. This project aims to address the following major problems:

Existing research has been limited to the "Mathematical vs. Physical" Data Gap and extensively used SMOTE [6, 7] to address imbalanced data. The problem with SMOTE, however, is that it also draws lines between the existing data points, and thus it generates "mathematically average" synthetic data. In air quality, the rises of pollution are not only given by averages but by complex air physics. In contrast, these standard methods yield a "noisy" data set that does not accurately represent the hazard, and thus do not provide an opportunity to investigate physics-based generative models such as CTGAN.

Other imputation methods are used for filling up sensor gaps, like KNN or Median based imputation [7,11]. All of these methods are "static" and do not take into consideration the fast time-varying nature of wind and traffic. These traditional methods would obscure the data by "erasing" the high pollution peaks that a forecasting system is supposed to identify after several hours of a sensor trip. There is a definite lack of a mechanism that will offer high accuracy and confidence score when data is missing.

Most of the high precision models from the literature [10,13] are black-box models. They issue a "Hazardous" alert, without any reason. An air quality official should have a way of knowing which pollutant is the problem that would need to be addressed (e.g., halt construction vs diverts traffic). But existing frameworks lack integrated Explainable AI (SHAP) that makes them hard to trust and implement by the authorities.

Limited Spatial Transferability Most studies are "city-locked" research, and models are developed and validated with data from a single city, for instance, in Chennai or Hyderabad [4,13]. The major research gap is to find out whether it is possible to transfer the OpenAQ-based model to local context in India without compromising the accuracy

of the model. There is no current research to demonstrate the robustness of these systems across geographic boundaries, and there is no Transferability Matrix available.

Chapter 3

Proposed Methodology and Implementation

3.1 Methodology

3.1.1 Phase 1: Data Preparation

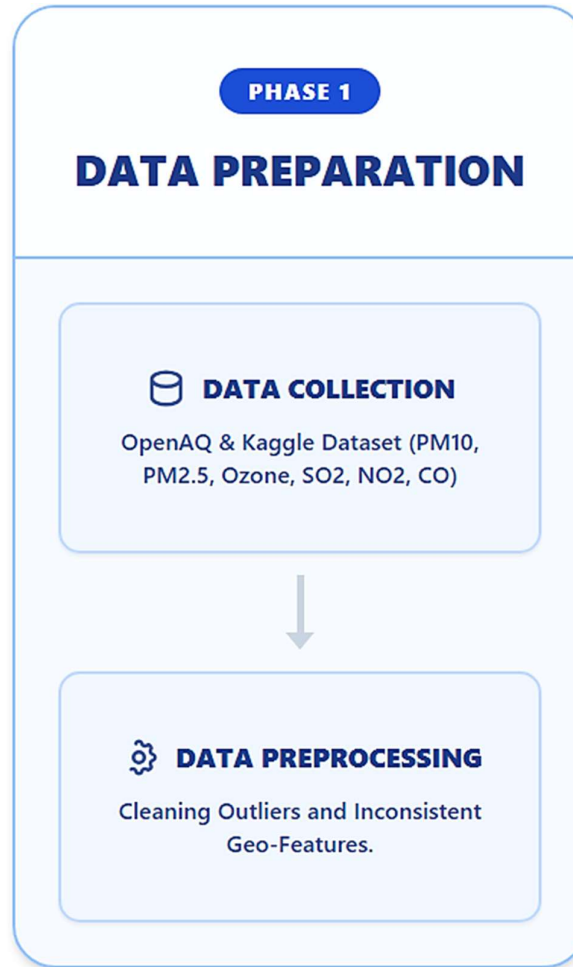


Figure 3.1: Data Preparation and Cleaning

The first step in the proposed methodology is completely devoted to data preparation. The raw historical OpenAQ data (2016-2022) [1] had to be pre-processed extensively to be mathematically and geographically consistent before being utilized for any exploratory data analysis (EDA) or machine learning models – either for the baseline or optimized ones. The Kaggle dataset for 2023-2025 [2] was collected in a clean, structured format and thus did not require any of the above mentioned preprocessing steps.

- Structural Formatting and Unpivoting:

The raw data was originally collected in a very disparate long format. It was then unpivoted and aggregated by unique identifiers (location, city, country and timestamp). As a result, it was expanded to a wide format with a separate column for each primary pollutant (PM2.5, PM10, NO2, SO2, CO and O3).

- Unit Standardization and Outlier Constraints:

A first audit found that 3.79% of the records were severe physical outliers, mostly because of the physical sensors recording the wrong units. In order to correct this but not to lose valuable data, a strict standardization process was carried out:

- Unit conversion: Pollutants incorrectly recorded in parts per million (ppm) were identified and changed to the standard microgram per cubic meter ($\mu\text{g}/\text{m}^3$) format.

This was achieved using the following conversion formula:

$$C_{\mu\text{g}/\text{m}^3} = \frac{C_{\text{ppm}} \times MW}{V_m} \times 1000$$

where,

$C_{\mu\text{g}/\text{m}^3}$ = Concentration in micrograms per cubic meter

C_{ppm} = Concentration in parts per million

MW = Molecular weight of the specific gaseous pollutant (g/mol)

V_m = Molar volume of an ideal gas ($\approx 24.45 \text{ L}$ at 25°C)

For example, CO concentrations lower than 50 were multiplied by a default CO conversion ratio of 1145. The molecular weight conversions for NO2, SO2, and O3 were similar.

- Applying AQI Constraints: Physical atmospheric limits for each pollutant were established (e.g. 0-1000 $\mu\text{g}/\text{m}^3$ for PM2.5 and 0-50000 $\mu\text{g}/\text{m}^3$ for CO). Values outside of these reasonable bounds were set to NaN rather than dropping them, because the rest of the pollutants may have good readings. No imputation has been made at this stage

since any missing values were deliberately left to be handled by the adaptive time-interpolation process in the subsequent stages.

- Validation of City and Country Records:

A filtration system was used to guarantee the integrity of the location information. The country and city columns were scanned using Regular expressions (Regex) that would find digits, special characters or non-alphanumeric symbols in the data, representing corrupted data. Such values represent, that records were lost from the data (irretrievable records or garbage records as in system test strings).

- Geospatial Correction with Geopy API:

One of the most significant problems that was found within the raw data was that the latitudes and longitudes were not being correctly entered as 0.0 or 1.0. As a result of these discrepancies, the geopy library and the Nominatim geocoder were added to address these issues. The API looked for the region names that were validated, and returned the actual latitude and longitude coordinates automatically. Those corrected coordinates were then added back in to fill in missing geospatial data gaps.

3.1.2 Phase 2: Baseline Workflow

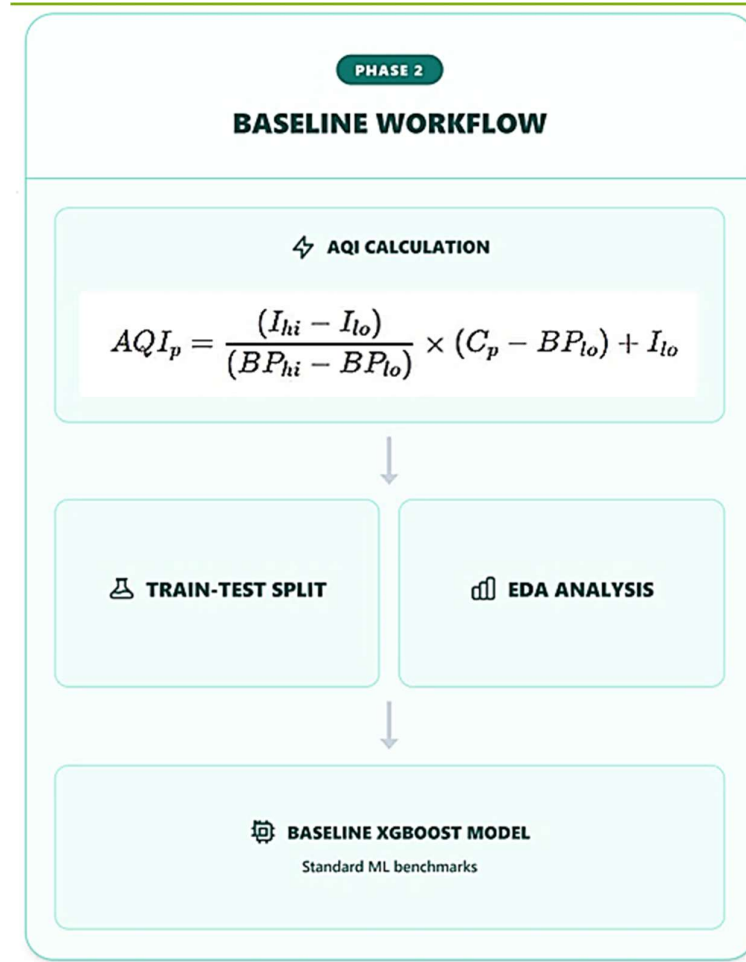


Figure 3.2: Baseline Pipeline and XGBoost Implementation

Phase 2 is based on the foundations laid in the previous phase, and aims to build the baseline performance, without any optimization. The cleaned data was then used in order to compute the basic air quality parameters, perform exploratory analysis and train a baseline machine learning model. This is used as a standard to evaluate the proposed optimized framework

- **AQI Calculation and Categorization:**

To quantify the pollution levels, the raw pollutant concentrations ($\mu\text{g}/\text{m}^3$) were converted into standard Air Quality Index (AQI) sub-indices. Also, the concentration breakpoints for both the datasets vary according to government standards. The calculation follows the standard piecewise linear interpolation formula for each specific pollutant p :

$$AQI_p = \frac{(I_{Hi} - I_{Lo})}{(BP_{Hi} - BP_{Lo})} \times (C_p - BP_{Lo}) + I_{Lo}$$

where,

I_p = The calculated AQI sub-index for pollutant p

C_p = The input concentration of pollutant p

BP_{Hi} = The concentration breakpoint that is $\geq C_p$

BP_{Lo} = The concentration breakpoint that is $\leq C_p$

I_{Hi} = The corresponding AQI value for BP_{Hi}

I_{Lo} = The corresponding AQI value for BP_{Lo}

A highly optimized, customized NumPy masking function was developed to calculate these sub-indices for all six primary pollutants (PM2.5, PM10, NO2, SO2, CO, and O3) simultaneously. Once all individual sub-indices were computed, the final overall AQI for a given temporal record was determined by taking the maximum sub-index:

$$Overall\ AQI = \max(I_{PM\ 2.5}, I_{PM10}, I_{NO2}, I_{SO2}, I_{CO}, I_{O3})$$

- Defining AQI Categories (Standard and Localized):

Following the calculation, the continuous AQI value was binned into distinct health-risk categories.

- Global (US-EPA Standard): Used for initial baseline training on global data. Categories include: 0–50 (Good), 51–100 (Moderate), 101–150 (Unhealthy for Sensitive Groups), 151–200 (Unhealthy), 201–300 (Very Unhealthy), and 301+ (Hazardous).
- Localized (India CPCB Standard): Specifically utilized for the localized Indian dataset analysis. Categories include: 0–50 (Good), 51–100 (Satisfactory), 101 – 200 (Moderate), 201 – 300 (Poor), 301 – 400 (Very Poor) and 401 and above (Severe).

- Exploratory Data Analysis (EDA):

An Exploratory Data Analysis was carried out before training the baseline model to gain insights into the feature dynamics:

- Linear Correlation: The purpose of it was to find out whether there is any collinearity between the individual pollutants, as well as whether there is any direct

association with the effects of each pollutant on the final AQI value, by the construction of a correlation matrix (using a Seaborn heatmap). This was quantified using the Pearson Correlation Coefficient:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

where,

r = Pearson Correlation Coefficient

x_i, y_i = Individual samples of the two variables being compared.

$\bar{x}, \bar{y}$ = The mean values of those variables

- Class Separability: Stratified sampling—with the random samples limited to a maximum of 500 from each class of AQI—was used to visualize the separate class AQI distributions (e.g., PM2.5 vs. NO2) to assess how well separated the class AQI distributions were in the feature space.
- Boxplots: To get the idea of how much the primary drivers, such as PM2.5, vary and how concentrated they are in each category of health risk, boxplots are created. The y-axis is indicated by a logarithmic scale (log) due to the high variability in the levels of pollution, so as to give a clear impression of the increase in the median pollutant concentration as the AQI category becomes more severe.

- Temporal Train-Test Splitting:

Air quality data is very time-dependent, and standard train-test split would result in data leakage: future data would affect predictions in the past. A strict temporal split was used to avoid this. The data set has been sorted chronological by the timestamp at which the data was recorded, and the first 85% of the data set is used for training, defined as the historical data set, the remaining 15% is used for testing, that is, the future prediction simulation data set.

- Baseline XGBoost Modeling and Evaluation:

To model the predictive power without any advanced data engineering or synthetic generation, a standard eXtreme Gradient Boosting (XGBoost) Regressor was implemented.

- Feature Selection: Only simple, raw features (geographic location – latitude, longitude – and unoptimized raw pollutant concentrations) were used to train the model.
- Training Details: The algorithm was trained using a histogram-based method (tree_method='hist') that ran with 1000 estimators, a maximum depth of 10 and learning rate of 0.05 for speed.
- Evaluation Strategy: RMSE and MAE were calculated for the model's prediction on the unseen 15% testing sets using following formulas:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where,

n = Total number of observations in the test set

y_i = The actual recorded AQI value

$\hat{y}_i$ = The predicted AQI value generated by the model

3.1.3 Phase 3: Proposed Optimized Framework

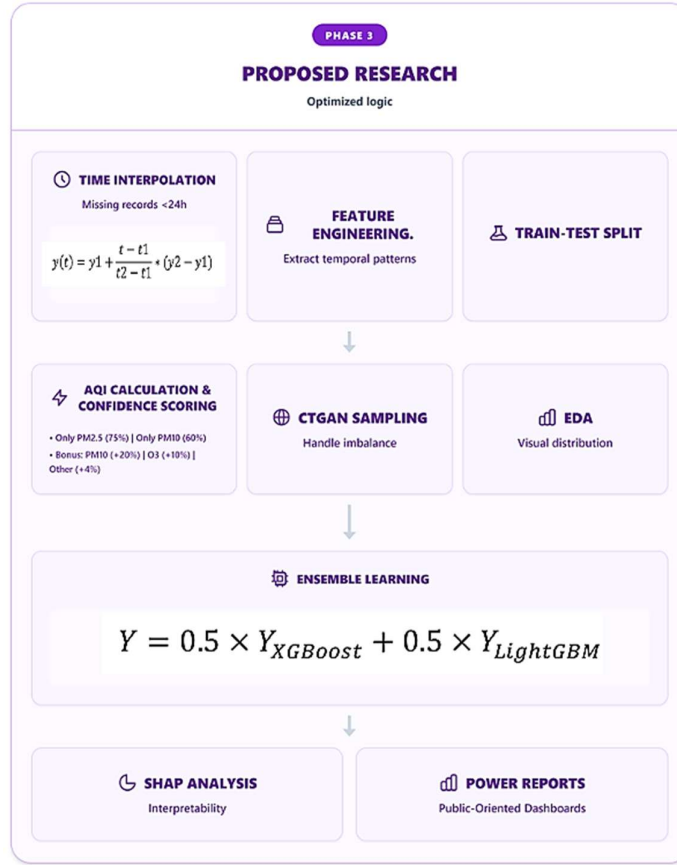


Figure 3.3: Proposed Research and Optimized Framework Pipeline

- Time Interpolation:

To resolve missing time records in the OpenAQ dataset due to temporary sensor failures, a linear time interpolation method has been used. The assumption behind this technique is that the rate of change in pollutant concentration between two known observations occurs at a constant rate. Hence, this method makes it applicable for short gaps of missing time series environmental data. A missing pollution concentration y at time t can be estimated by using the following equation to estimate the value:

$$y(t) = y1 + \frac{t - t1}{t2 - t1} * (y2 - y1)$$

$y(t)$: The interpolated pollutant concentration at the missing timestamp t .

$y1$: The actual recorded concentration at the nearest known timestamp before t

y_2 : The actual recorded concentration at the nearest known timestamp after t .

t : The target timestamp for which we need to estimate the value.

t_1 : The actual time corresponding to the reading y_1 .

t_2 : The actual time corresponding to the reading y_2 .

To limit the creation of fake trends or smoothing out extreme pollution events, we have only performed interpolation on short gaps of less than 24 hours. Gaps greater than 24 hours have been kept the same as their original values. Thus, through this type of interpolation, we estimated an additional 4,951,806 missing hours, giving us a total of 7,372,206 records in OpenAQ dataset for training the model.

- Feature Engineering:

Additional variables were introduced to help the model detect and fit to more complex patterns. The new features that were developed are `local_hour`, `local_month`, `local_day_of_week`, `is_weekend`, `season`, `time_of_day`, `PM2.5_24h_mean`, `PM10_24h_mean`, `NO2_24h_mean`, `SO2_24h_mean`, `O3_24h_mean`, `CO_24h_mean`, `hour_sin`, and `hour_cos`. The model is more likely to detect and fit the time-based trends if temporal components are extracted from the timestamp column. The model can use the `is_weekend` variable to identify differences between weekday and weekend pollution levels, as pollution levels are significantly higher on weekdays than on weekends. Rolling mean calculated over 24-hour periods for each of the major pollutant types also allows the model to develop a more accurate representation of short-term temporal trends in pollutant concentration.

- AQI Formulation and Confidence Scoring:

(Note: The base AQI calculation and standard categorization steps in this phase are mathematically identical to the baseline workflow established in Phase 2).

To handle missing pollutant values, a confidence score mechanism was introduced. In the proposed approach, greater importance is given to PM2.5 and PM10, as they are the dominant pollutants in most urban air pollution cases. The base score is determined first: if PM2.5 is present, the baseline confidence is set to 75%; if only PM10 is present, the baseline confidence is 60%; and if neither PM2.5 nor PM10 is present, the baseline confidence drops to 35%. Additional bonus scores are added when other pollutants are

available. Specifically, +10% is added if PM10 is also present, +5% for Ozone (O₃), and +4% for each of the minor gases (NO₂, SO₂, and CO).

- Minority Sampling using CTGAN:

Compared to basic techniques such as random oversampling or SMOTE, CTGAN is more advanced because it understands the distribution of tabular data and can produce more realistic synthetic samples. It works by training a generative adversarial network where a generator generates synthetic records while a discriminator tests their authenticity. This training process follows the GAN Min-Max Objective Function:

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

where,

$V(D, G)$: The value function of minimax game between Generator & Discriminator.

$D(x)$: The probability that the real data x is "authentic" rather than synthetic.

$G(z)$: The synthetic record produced by the generator from random noise z .

E : The expected value over the data distributions.

This adversarial framework allows the model to learn complex relationships between features and improving model learning. allowing the model to learn complex relationships between features and improving model learning. The number of samples in the “Very Unhealthy” category increased from 25,611 to 81,798, and for the “Hazardous” category, it increased from 13,185 to 84,146 in OpenAQ dataset. Furthermore, for Kaggle data, the number of rows for ‘Poor’, ‘Very Poor’ and ‘Severe’ increased by 48,710, 27,464 and 7924.

- Train-Test Split:

The temporal 85/15 train-test split criteria remains strictly consistent with the methodology used in the baseline workflow.

- Ensemble Learning:

Two gradient boosted systems integrated as part of ensemble learning were XGBoost & LightGBM. The models used were trained using a learning rate of 0.05 and maximum

tree depth of 10, using 1,000 as the maximum number of estimators. Overfitting prevention was achieved using early stopping mechanisms. Finally, predictions from both models were averaged together with an ensemble method:

$$Y = 0.5 \times Y_{XGBoost} + 0.5 \times Y_{LightGBM}$$

- Final Interpretability and Deployment

To ensure the ensemble model remains interpretable and transparent, **SHAP analysis** was utilized. It assigns each feature an importance value for a particular prediction by calculating the contribution of that feature across all possible combinations of features. The Shapley value φ for a feature i is defined as:

$$\frac{|S|! (n - |S| - 1)!}{n!} [f(S \cup \{i\}) - f(S)], \quad S \subseteq N \setminus \{i\}$$

where,

n = The total number of features.

S = A subset of features excluding feature i

$f(S)$ = The model's prediction using only the features in subset S

$f(S \cup \{i\})$ = The prediction after adding feature i .

Also, the deployment of Power Reports was been made to translate the complex ensemble predictions into public-oriented dashboards.

3.1.4 Phase 4: Evaluation and Local Analysis

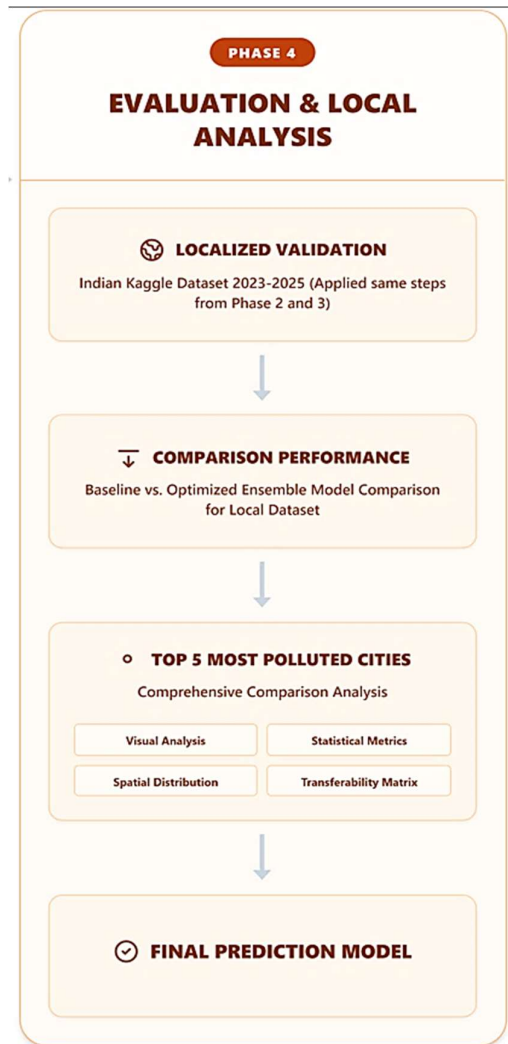


Figure 3.4: Evaluation and Local Analysis Workflow

The last step of the methodology is to thoroughly test and validate the proposed framework. The model is tested on a completely new, local dataset to ensure that the model is not only well-fitting the historical global information, but also robust.

- **Localized Validation:**

The model was then validated with a localised Indian Kaggle dataset of recent air quality data from 2023 to 2025 to gauge its adaptability in real-world scenarios. This new data set didn't need any Data Preparation or preprocessing pipeline (outlier removal, unit conversion, geographical inconsistencies) used for the OpenAQ data set in Phase 1, because it was already collected consistently.

- Performance Comparison:

After local data was created, the actual head-to-head performance was conducted. In both Phase 2 and Phase 3, both the models were asked to predict the AQI on this new local data, the Baseline XGBoost model and the proposed Optimized Ensemble model. This comparison demonstrates the quantitative gains (evaluated by common error metrics such as RMSE and MAE) from the advanced feature engineering and synthetic sampling methods proposed in the framework.

- Top 5 Most Polluted Cities Analysis:

An extensive and detailed study was made on the top 5 cities in the localized dataset rather than considering the overall figures. The deep dive is organized in four pillars:

- Visual Distribution Analysis: Boxplots were used to visualize the distribution of AQI across top cities, in comparison to satisfactory (100) and severe (300) thresholds, showing the median, interquartile range (spread), and extreme pollution spike days (outliers).
- Air Quality Composition: Stacked bar charts were used to show the percentage of time each city spent in each health category of the AQI (from 'Good' to 'Severe') to show the proportion of time breathable and hazardous air was experienced during the year.
- Statistical Metrics: A hard statistical summary was calculated for each city, including the mean, median, maximum values of AQI and the standard deviation which quantifies the severity and volatility of local pollution.
- Cross city transferability matrix: It was drafted to assess the extent of transferability of model patterns learned in various localized climates. The proposed ensemble, XGBoost and LightGBM, was systematically trained with 75% of the data from one city and tested with unseen data from all other cities, and then the resulting RMSE and MAE matrix was used to demonstrate the robustness of the geographical data.

- Final Prediction Model:

This stage is to confirm the methodology yields a highly accurate, interpretable and transferable tool that can be used for environmental monitoring.

3.2 Simulation Environment

Technical infrastructure, software tools and specific datasets used to develop, train and validate proposed air quality forecasting framework:

3.2.1 Software Configuration

The development environment and software stack were chosen to provide both high-speed data processing and accurate geospatial mapping, while also being transparent to enable model explanation.

- **Primary Development Platform:**
The main computational platform was Primary Development Platform: Google Colab, which offers high-performance CPU and RAM resources to process millions of records and train memory-hungry models such as CTGAN.
- **Data Manipulation and Numerical Computing Libraries:**
 - Pandas: The key tool for data structuring, unpivoting and time-series indexing.
 - NumPy: For fast vectorized math operations, especially for computing fast AQI sub-indices.
- **Geospatial Processing Library:**
 - Geopy: Automated geocoding using Nominatim API, correcting latitude and longitude coordinates that are broken by searching in the region for its name.
- **Machine Learning and Synthetic Sampling Libraries:**
 - Scikit-learn: Used for baseline modeling, temporal train-test splitting, and calculating error metrics (RMSE/MAE).
 - XGBoost & LightGBM: Implemented as the key elements of the proposed ensemble approach for high accuracy regression.
 - CTGAN: A specific generative adversarial network to deal with class imbalance by generating realistic data in "Very Unhealthy" and "Hazardous".

- **Model Interpretability Library:**
 - SHAP: Introduced to address the “black-box” problem of the ensemble model, with explanations at both a local and global level of the most significant pollutants for each individual prediction of the AQI.
- **Visualisation and Reporting:**
 - Power BI: Used to develop public facing dashboards and "Power Reports" to enable complex model outputs to be translated into user-friendly, City level messages.
 - Seaborn & Matplotlib: Applied to internal Exploratory Data Analysis (EDA) from boxplots to heatmaps and temporal trend visualization.

3.2.2 Global Dataset Description (OpenAQ)

The first baseline model and historical training have been carried out with data from OpenAQ [1]. OpenAQ collects and standardises data from a wide variety of sources around the world from government monitoring agencies, research organisations and verified, low-cost sensor networks to give a single picture of air quality throughout the world.

- Scope: They span a long time from 2016 to 2022.
- Data Volume: The original OpenAQ data consisted of 2,431,075 rows and 14 columns.
- Role in Project: To supply the huge amount of data required to set up a global baseline and the relationships between various pollutants and different international topographies on which the model was trained.

Definitions for features and pollutants:

- timestamp: Records the precise date and time of reading in UTC, which is important for understanding the daily patterns and seasonal cycles.
- country / region_name: Categorical data that denotes the larger administrative areas to be used for analysing regional pollution trends.
- station_name: A unique identifier for the particular sensor site which can be used for learning site-specific behaviour.

- latitude / longitude: Accurate geospatial location information for the mapping of pollution hotspots and distance calculations from sensor to sensor.
- source_name: Contains information defining the agency or network provider (such as "OpenAQ") that provides the data, ensuring its provenance and traceability.
- PM2.5: Fine particles ($<2.5\mu\text{m}$) that enter the blood stream ($\mu\text{g}/\text{m}^3$), main parameter for health risk assessment.
- PM10: Fine particles ($<10\mu\text{m}$) which are usually from the road and building dust.
- NO2 (Nitrogen Dioxide): A gaseous indicator of combustion at high temperatures, mainly as an indicator for the intensity of traffic by cars.
- SO2 (Sulfur dioxide): An indicator of industrial activity, mainly from burning of coal and petroleum.
- O3 (Ozone): Secondary pollutant which is produced by chemical reactions in the presence of sunlight, important for midday smog prediction.
- CO (Carbon Monoxide): A chemical produced by incomplete burning of fuel that is toxic and highly associated with heavy traffic.

3.2.3 Localized Dataset Description (India Kaggle Dataset):

The secondary data was sourced from Kaggle (contributed by Saikiran Udayana) for localized testing and analysis for “Top 5 Most Polluted Cities” analysis [2]. This data was used as the "unseen" environment to check how robust the trained model was.

- Time Period: 2023-2025, modern and relevant perspective on air quality situation in India.
- Data Volume: Has a total of 842,160 records available that can be used for granular, city-level performance evaluation.
- Structure: This dataset is structured to be consistent with the OpenAQ dataset, with the same primary pollutant's concentrations (PM2.5, PM10, NO2, SO2, CO, O3), geospatial coordinates, and temporal timestamps. This meant there was no need for additional structural re-engineering to be done to utilize the pre-trained pipeline.
- Role in Project: This data set was very important for testing the model transferability. It helped in focused analysis of the 5 target cities (Gurugram, Delhi, Patna, Lucknow, Kolkata).

Chapter 4

Results

4.1 Outcomes of the Proposed Model

The potential performance of the proposed framework was initially assessed with reference to a global baseline (derived from the OpenAQ dataset). This phase is used to compare the unoptimized XGBoost model (Baseline) with the optimized individual models and the optimized final weighted ensemble. The goal was to measure the error reduction obtained by adding the CTGAN-based oversampling and the powerful temporal feature engineering.

Table 4.1: Performance Metrics Comparison for the OpenAQ (Global) Dataset

Model Category	Model Type	RMSE	MAE	% Improvement (RMSE)	% Improvement (MAE)
Baseline	XGBoost	6.84	0.83		
Optimized	XGBoost	4.36	0.62	36.26%	25.30%
Optimized	LightGBM	3.98	0.88	41.81%	-6.02%
Final Proposed	Ensemble	3.77	0.60	44.88%	27.71%

The test of the regional reliability and transferability of the framework was carried out using a localised Indian data set with high density records from major urban centres. This assessment shows the ability of the enriched ensemble to cope with the Indian subcontinent's distinct pollution episodes and extreme weather conditions throughout the seasons.

Table 4.2: Performance Metrics Comparison for the Kaggle (India) Dataset

Model Category	Model Type	RMSE	MAE	% Improvement (RMSE)	% Improvement (MAE)
Baseline	XGBoost	6.24	1.04		
Optimized	XGBoost	6.49	1.20	-4.01%	-15.38%
Optimized	LightGBM	5.21	1.18	16.51%	-13.46%
Final Proposed	Ensemble	5.39	1.02	13.62%	1.92%

The key findings from this comparison include:

- **Model Resilience in High Volatility:** The Indian dataset (Kaggle) is characterized by higher pollution peaks and seasonal spikes. While the optimized XGBoost struggled with this increased variance (showing a slight drop in baseline performance), the LightGBM model excelled, achieving a 16.51% improvement in RMSE. This suggests that LightGBM's leaf-wise growth strategy is more effective at capturing the "extreme" pollution events common in Indian cities.
- **The Stabilizing Effect of the Ensemble:** The Final Ensemble emerged as the most reliable predictor across both datasets. In the Indian context, the ensemble was the only model to maintain a positive improvement in both RMSE (13.62%) and MAE (1.92%). By balancing LightGBM's aggressive peak-fitting with the stability of the ensemble weights, it smoothed out the prediction errors that individual models could not resolve alone.
- **Generalization vs. Specialization:** The framework achieved a total 44.88% RMSE improvement globally and a 13.62% improvement in India. The lower percentage in the Indian dataset is a reflection of the localized data's higher complexity; however, the fact that the ensemble consistently outperformed the baseline in both environments validates the transferability of the proposed architecture.

Table 4.3: Cross-Transferability Matrix (RMSE) for Top 5 Polluted Indian Cities

	Gurugram	Delhi	Patna	Lucknow	Kolkata
Gurugram	11.76	10.01	7.89	6.56	8.55
Delhi	73.23	44.19	6.33	11.57	9.31
Patna	118.51	87.97	7.93	28.82	7.12
Lucknow	107.27	75.67	7.07	21.08	6.76
Kolkata	120.91	90.45	8.37	31.33	7.08

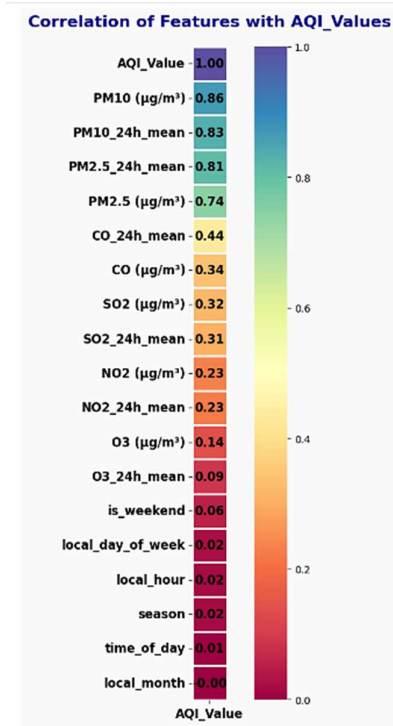
- **The Gurugram Anomaly:** The matrix also brings out that Gurugram is the worst place to predict with the external models. The RMSE value skyrockets to more than 100 when models trained on the cities of Patna, Lucknow or Kolkata make predictions for the air quality of Gurugram. This indicates that models trained in other cities are not suitable for the city of Gurgaon because there are pollution

drivers unique to the city with high volatility (probably heavy industrial or construction dust).

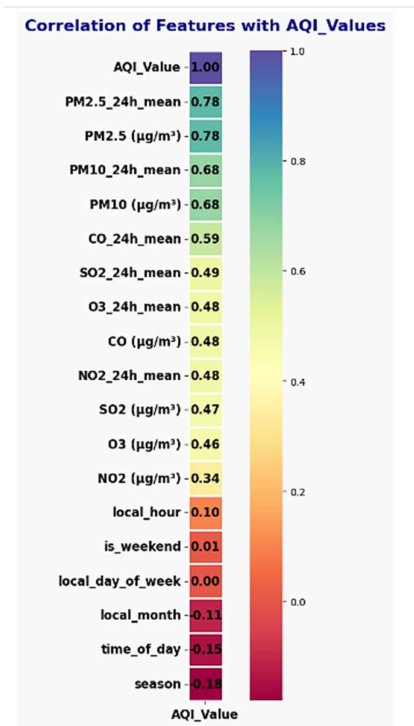
- Predictive Stability in Patna and Kolkata: Intriguingly, Patna and Kolkata demonstrate very low values of RMSE (between 6.33 and 9.31) irrespective of the city model. This means that the pollution pattern in these cities is more "stable" or "traditional" and is easily recognizable by any model which has been trained on Indian urban data.
- Model Specialization (The Diagonal): While the best option would be self-prediction (the diagonal), the Delhi trained model has an extremely high RMSE of 44.19 with itself data as compared to Patna data with RMSE of 6.33. This illustrates that the Delhi data set may show extreme "Hazardous" events which may be very difficult to model even in the Delhi region, whereas the "simpler" patterns in Patna are comparatively easier to predict using the model.

4.2 Result Analysis

4.2.1 Exploratory Data Analysis Plots



*Figure 4.1: Feature Correlation Heatmap
For Optimized Global Dataset*



*Figure 4.2: Feature Correlation
Heatmap for Optimized Indian Dataset*

- Linear Correlation Matrix:
 - The two heatmaps show that the major drivers of the AQI are PM2.5, PM10 with their 24h_mean features have a higher correlation than other pollutants and features.
 - The Indian dataset has a much wider multi-pollutant dependency with much higher correlations with CO (0.47 - 0.59), SO2 (0.55 - 0.71), and O3 (0.52 - 0.66) as compared to the Global dataset.
 - A pronounced regional difference is seen in the temporal features – the Indian data set exhibits a much stronger response to seasonality and time of day indicating the presence of more significant pollution cycles in this region than in the Global data.
 - Together, these correlations make the ensemble's design reasonable, because it has to handle high multi-collinearity among pollutants and have to retain accuracy by taking advantage of local temporal characteristics.

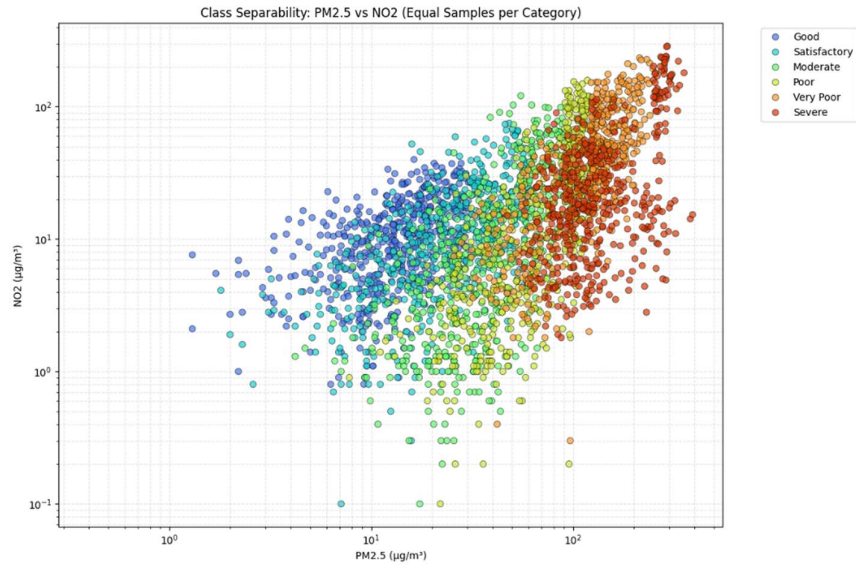


Figure 4.3: Class Separability for Baseline Indian Dataset

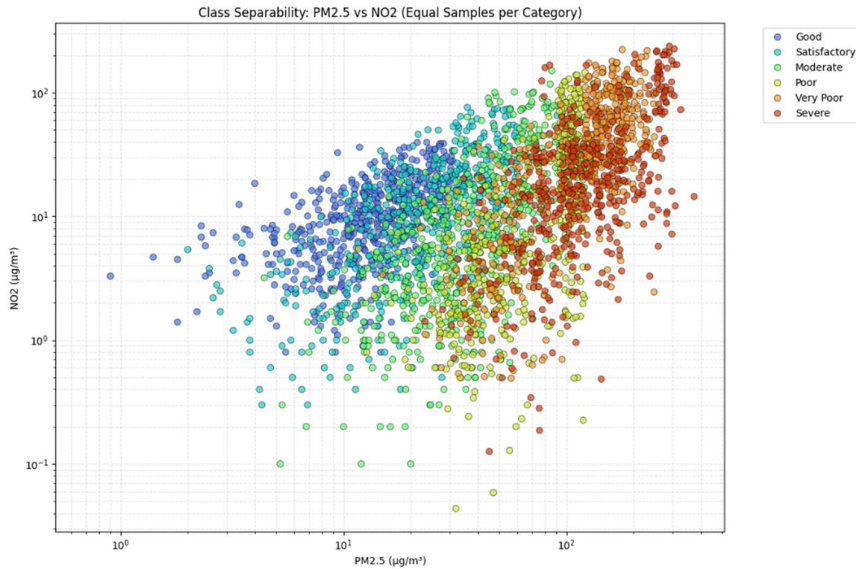


Figure 4.4: Class Separability for Optimized Indian Framework

- Class Separability
 - The Baseline plot (Fig. 4.3) shows that there is a lot of category overlap, especially in the mid-range (10 to 100 $\mu\text{g}/\text{m}^3$); the categories 'Satisfactory' and 'Moderate' are intermixed making it more difficult to define the decision boundaries for the regressor.
 - In the Optimized framework (Fig. 4.4), the clusters for classes 'Very Poor' and 'Severe' are clearly thicker. This is a consequence of the oversampling method used in CTGAN, which enabled a better representation of minority high pollution events.

- By calculating 24-hour rolling averages, the Optimized version of the feature space is more stable (there is less noise from the sensor), bringing data points that are situated far away from the category centroids back into the zone of influence.
- This greater separability, particularly at the pollutant extremes, is the main reason for Ensemble Model's better performance in reducing the MAE and correctly capturing hazardous air quality changes.

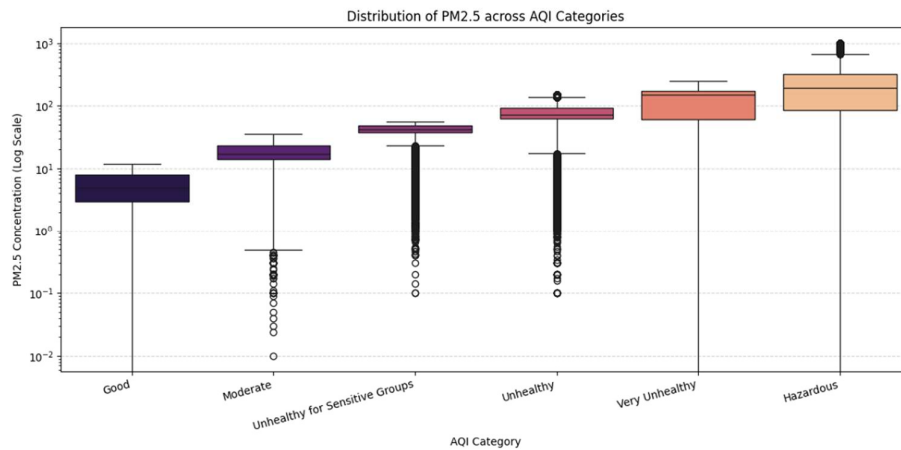


Figure 4.5: Distribution of PM2.5 across AQI Categories (OpenAQ - Global)

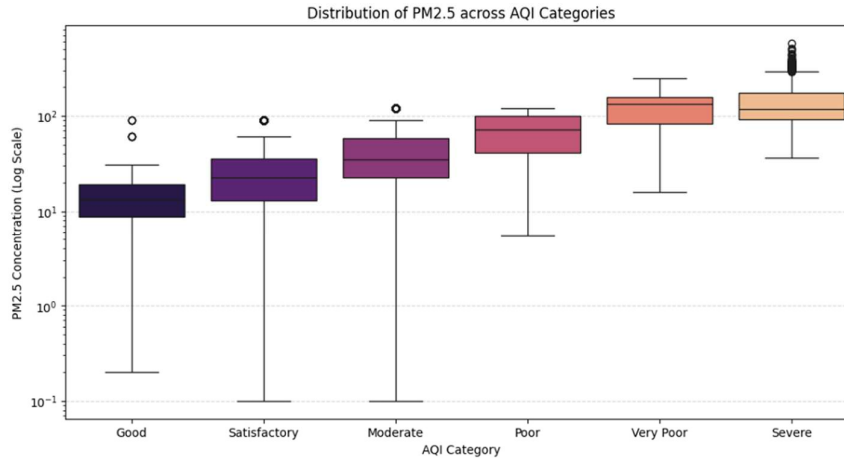


Figure 4.6: Distribution of PM2.5 across AQI Categories (Kaggle - India)

- Box Plots
 - The interquartile ranges are relatively small (Fig. 4.5), "Moderate" and "Unhealthy" ranges are especially small for the OpenAQ Global data. Compare that with the Indian Kaggle plot (Fig. 4.6), where the boxes are much wider for each class AQI, with a large spread of pollution levels for each class.

- An important thing to observe is the density of outlier is that the Indian data exhibits substantial "upward" outliers even in the Satisfactory and Moderate classes with PM2.5 air quality values occasionally reaching values typical of the "Poor" class.
- The categories in the Global data are mathematically "cleaner", there is a smaller overlap between the whiskers of adjacent classes. The Indian data has a lot of overlap, which indicates that the local urban factors make it difficult to distinguish between categories without Feature Engineering (rolling means) that we applied in our optimized framework.

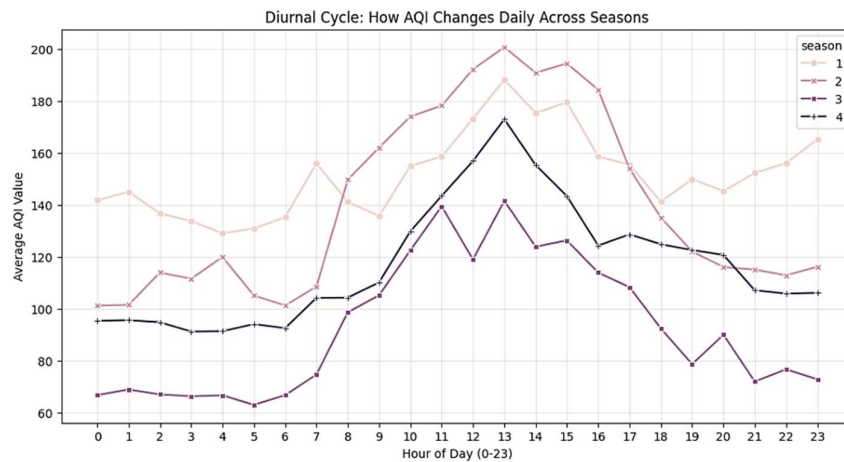


Figure 4.7: Diurnal AQI Cycles Across Seasons (Indian Kaggle Dataset)

- Diurnal Cycle Plot
 - The plot shows a noticeable "twin-peak" pattern in nearly every season. The first spike is from 7:00 AM until 10:00 AM (morning rush hour), and a second, worse and longer spike starts after 6:00 PM, reaching its peak around midnight. This study indicates traffic and boundary layer changes are the main factors affecting AQI in Indian cities.
 - The average AQI value is generally the highest for Season 2 (Post-Monsoon/Winter) and is often above 160, and the lowest for Season 3 (Monsoon), possibly due to pollutant washout through the rain in this season.
 - There is a sudden drop in AQI at midday in every season. This is an important physical consideration: when the sun starts to come out, the ABL is raised, causing a dispersion of pollutants and a brief reduction in ground level concentrations.

4.2.2 SHAP Analysis

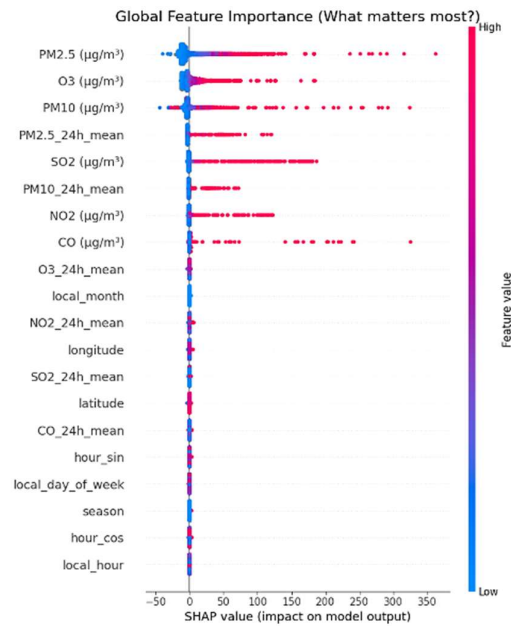


Figure 4.8: SHAP Beeswarm Plot for Optimized OpenAQ (Global) Model

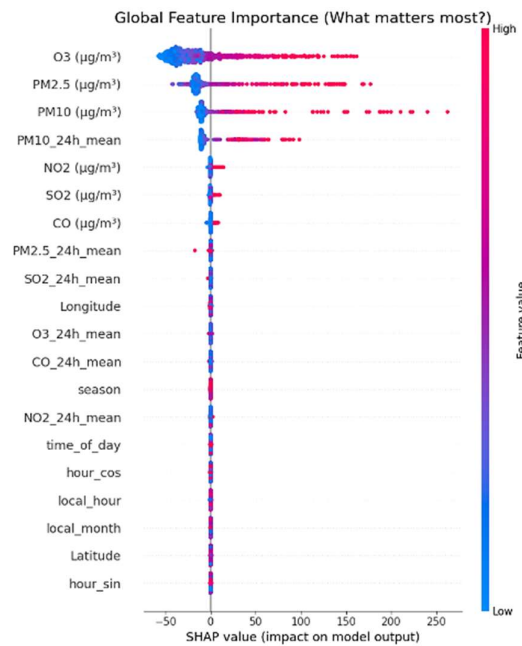


Figure 4.9: SHAP Beeswarm Plot for Optimized Indian Kaggle Model

- Beeswarm Plot
 - The model with Global Primary Drivers (Figure 4.8) shows that PM2.5 has the largest SHAP values and is the absolute big player. The Indian model (Fig. 4.9) is not so different from the global model, except that O3 (Ozone) is also a dominant

contributor to the final AQI prediction, with a much wider impact than it does on the global model.

- The Indian model is more sensitive to NO₂ and SO₂ than the Global model in the order of the pollutants in the feature ranking. The Indian framework indicates that the chemical mixture is capturing the complex mixture of chemicals found in Indian urban areas, and that nitrogen and sulfur oxides are more important in driving daily AQI variations in these areas.
- The 24h_mean feature is among the top 6 features used to build rank list in both data sets. Finally, the values of SHAP for PM_{2.5} 24h_mean and PM₁₀ 24h_mean, demonstrate that our model strongly depends on the temporal context, and not merely on the peaks of pollution occurring at given sensor outputs, which can be noisy.
- The bottom three features (longitude, latitude, and season) have relatively low SHAP values, indicating that this model uses them to “fine-tune” its predictions.

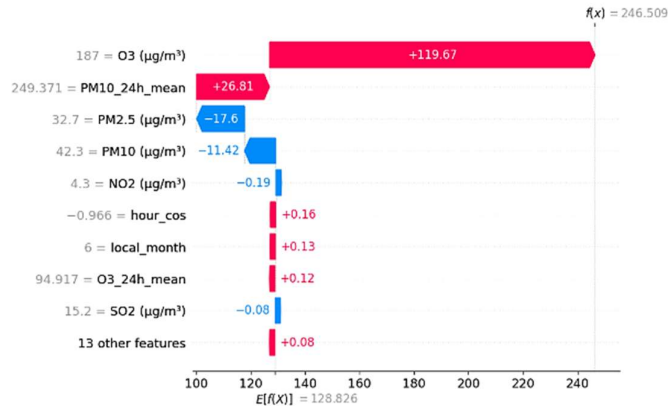


Figure 4.10: SHAP Waterfall Plot for a High-Pollution Instance (Indian Dataset)

- Waterfall Plot
 - The graph shows how the base value ($E[f(x)] = 128.8$) is changing to the end value ($f(x) = 246.5$). This "local" explanation shows that the model is not simply a guess, but rather a mathematical consideration of each feature for each particular hour of the day.
 - In this particular scenario, the major contributor is Ozone (O₃) which contributes to an increase in AQI by +119.67. This shows the framework's capability to detect "Ozone-dominant" pollution episodes, which are prevalent during high temperature afternoons in Indian cities even in the presence of relatively stable PM.

- Interestingly, the engineered PM10 24h_mean is going to counteract the pull-down effect that is being imparted by the instantaneous PM10 reading (+26.81). This confirms that you're correct with your feature engineering: The model knows that the last 24 hours were polluted, and is not going to underestimate the AQI due to a single spike in the sensor data.

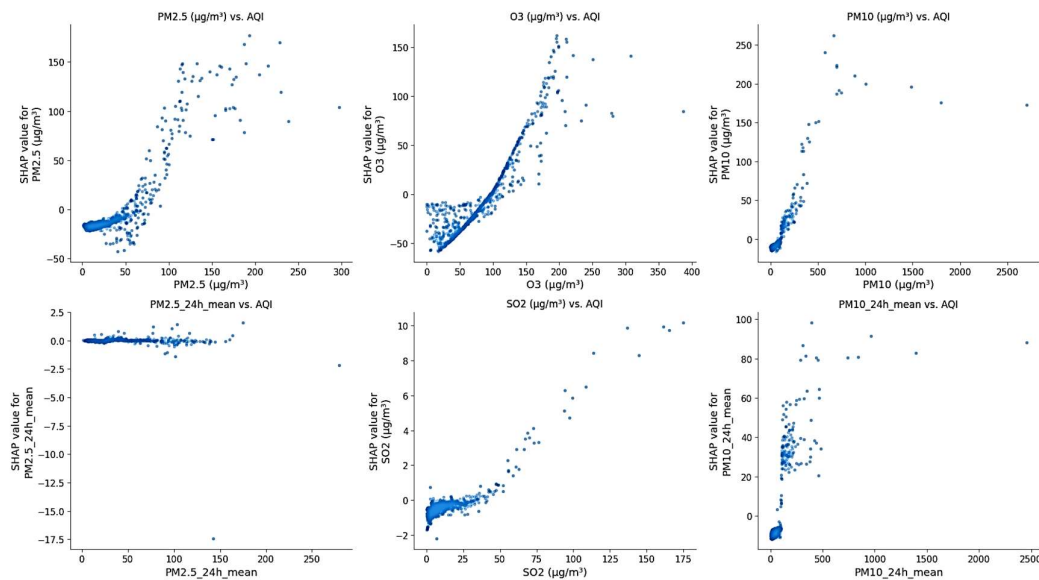


Figure 4.11: SHAP Dependence Plots for Key Pollutant Drivers (Indian Dataset)

- Dependence Plot
 - All of the major pollutants (PM2.5, O3, PM10) have a distinct “elbow” or inflection point. For example, in the O3 plot, SHAP value (impact on AQI) does not significantly change until around 100 $\mu\text{g}/\text{m}^3$, and then increases in an exponential manner.
 - PM2.5 and PM10 have a very linear-to-logarithmic relationship with the model output. The model has learned from the data to be quite sensitive to particulate matters at low concentrations, giving a substantial increase in the predicted AQI even with small increases in PM2.5.
 - The dependence plots of PM2.5 24h_mean and PM10 24h_mean are similar to that of the instantaneous values, but exhibit more stabilized "bands" of SHAP values.

4.2.3 Power BI Dashboards

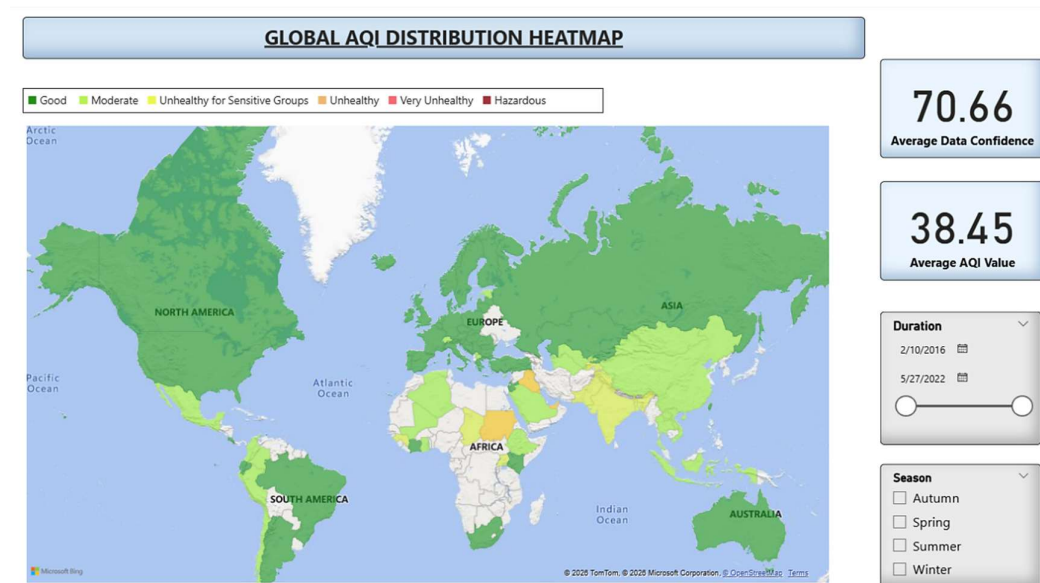


Figure 4.12: Global AQI Distribution Heatmap and Analytical Dashboard

- Real-time Visualization & Global AQI Dashboard
 - Geospatial Risk Mapping: The dashboard is equipped with an extensive Global AQI Distribution Heatmap, which uses a standard colour coded legend (from 'Good' to 'Hazardous', from green to dark red) to show the air quality status of the air across continents at a first glance. Beyond the map, the interface displays key metrics in real-time, such as Average Data Confidence (38.45%) and Average AQI Value (70.66%), providing a snapshot of air health around the world and the reliability of the sensors.
 - Temporal and Interactive Filtering: The dashboard is designed to be used in deep-dive analysis: It lets users choose a Season (Fall, Spring, Summer, Winter) or pick a Duration slider (from 2016 to 2022).
 - Granular Drill-Down Capability: The primary view is global, but there is a city-level drill down capability as well. This will enable stakeholders to transition from the continental level to specific cities and apply the tool to support urban policy-making and health communications.



Figure 4.13: Monthly Air Quality Trends and Pollutant Correlation Dashboard

- **Monthly Air Quality Trends**
 - **Seasonal Pollutant Dynamics:** The most important element of this view is the "Yearly Cycle: Dust (PM10) vs. Traffic Fumes (NO2)" chart, which compares the average of PM10 with NO2. This enables a better understanding of the interactions between various pollution sources (winter heating, summer dust etc) throughout the four seasons.
 - **Regional Monthly Heatmap:** The dashboard features a detailed Monthly Air Quality Index Heatmap by Region that shows how the AQI has changed over the course of the year in specific cities (such as Abruzzese, Accra, Adana). This table is important to indicate what constitutes a "pollution season" in each geographic area, which further supports the need for the seasonal features in the machine learning model.
 - **Aggregated Performance Metrics:** The AQI Peak Level gauge will give a global view of air quality and currently displays an average of 38.45, corresponding to the 'Good' - 'Moderate' categories on the standard scale.

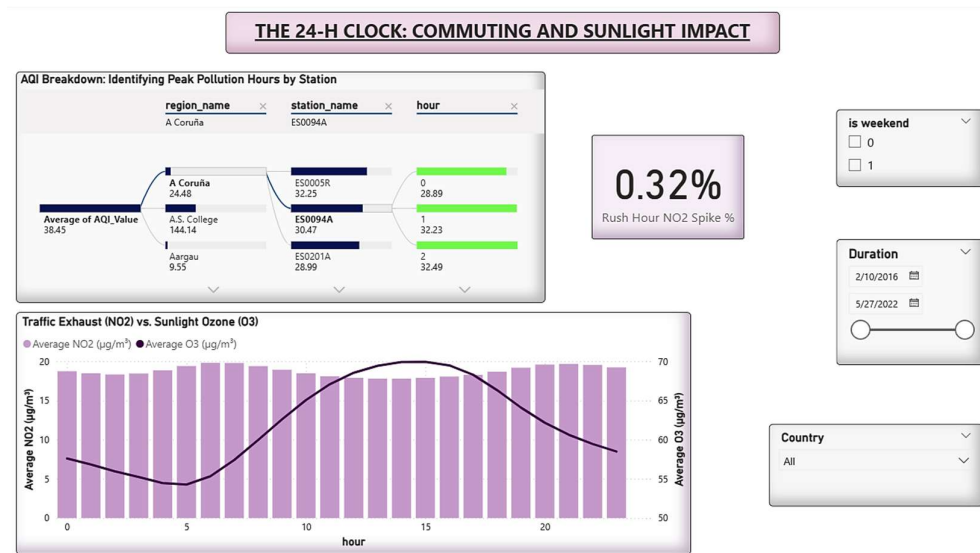


Figure 4.14: Impact of Sunlight Exposure and 24-Hour Cycle on AQI

- Sunlight Exposure and Commuting AQI Correlation
 - The "Traffic Exhaust (NO2) vs. Sunlight Ozone (O3)": This chart illustrates the inverse relationship between NO2 and O3, giving a visual confirmation of the photochemical nature of air pollution. NO2, which is from traffic exhaust, is fairly steady, with minor peaks, while O3 has a very distinct bell-shaped curve, with its peak occurring at the peak time for sunlight (roughly between 2:00 PM and 4:00 PM).
 - Quantifying Traffic Impact: The dashboard will also feature a dedicated metric, the "Rush Hour NO2 Spike %" (currently 0.32%), which specifically measures the spike in the amount of NO2 during regular commuting hours. This establishes a baseline of the chemical load from the city's activity (commuters) which can then be used to alter the weightings for the model according to whether it is a "Rush Hour" or "Off-Peak" time.
 - Hierarchical Peak Hour Identification: The system can be used to discover which hour is most dangerous to which station in detail, using the decomposition tree of the "AQI Breakdown". The layered organization (regional station hourly) allows stakeholders to identify the time of day during which a particular station (e.g., ES0094A) will see the most hazardous levels in order to offer the local context to the public health warning.

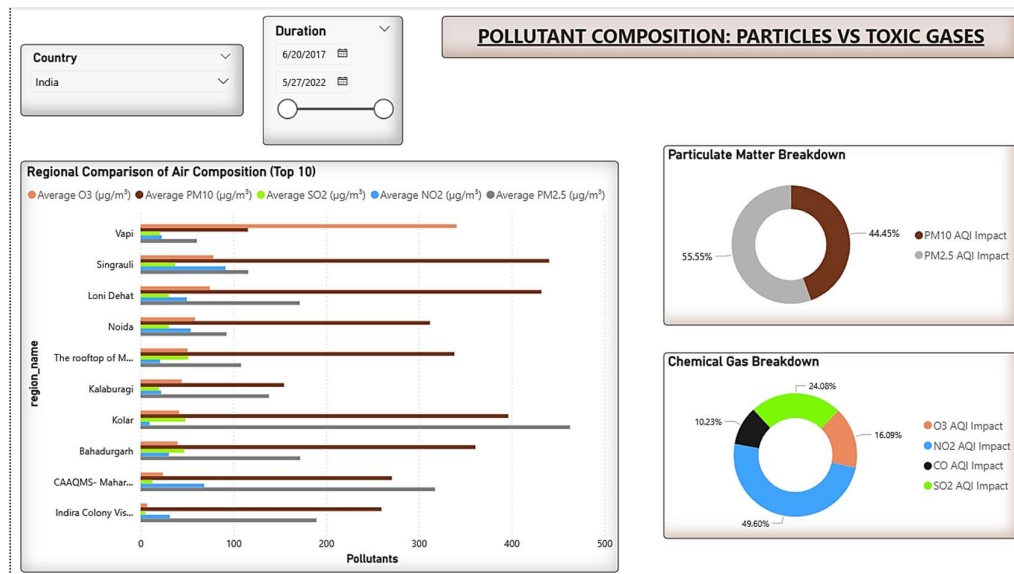


Figure 4.15: Comparative Breakdown of Particulates vs. Toxic Gases in Indian Cities

- Pollutant Composition and Source Partitioning (India Analysis)
 - Regional Chemical Fingerprinting: Pollution profiles are clearly observed at the top industrial cities of India by using the bar chart, "Regional Comparison of Air Composition". The cities of Vapi and Singrauli, for example, have significant amount of PM10, which is probably associated with industrial dust, while Kolar has a greater proportion of PM2.5.
 - Particulate Matter Dominance: The "Particulate Matter Breakdown" donut chart shows that PM2.5 contributes 55.55% of the particulate impact as compared to 44.45% by PM10. The SHAP importance plots presented earlier also indicate that fine inhalable particles are the main contributors to the "Severe" AQI levels predicted by the model; this data-driven insight confirms this.
 - Gaseous Driver Identification (NO2 Focus): The dashboard shows that in the "Chemical Gas Breakdown", the dominant toxic gas is NO2 accounting for 49.60%. This high percentage is followed by SO2 at 24.08%, reflecting the relevant contribution of the emissions from vehicles and from coal combustion in the Indian data set.

4.2.4 Top 5 Polluted Cities Analysis

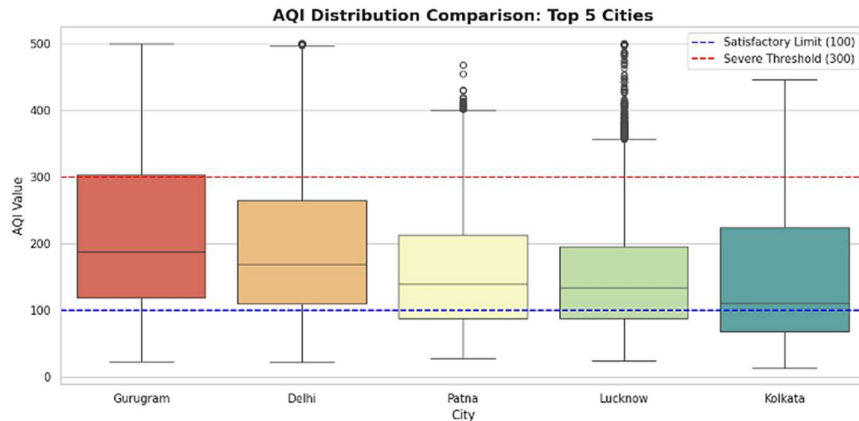


Figure 4.16: AQI Distribution for India's Most Polluted Urban Centers

- Box Plot:
 - The box plot shows that for all five cities (Gurugram, Delhi, Patna, Lucknow, and Kolkata) the median value of the AQI is consistently above the "Satisfactory Limit" (blue dashed line), which is 100. Interestingly, in Gurugram and Delhi the entire interquartile range (the box) is almost entirely above this limit, which means that 'Satisfactory' air quality is a rare statistical outlier and not the average.
 - Delhi and Lucknow have very high spikes in their data with the individual outliers on the maximum scale of 500. The black data points located above the whiskers are extreme pollution events which the model has to capture.
 - Gurugram has the highest 'typical' pollution levels with its 75th percentile crossing the 'Severe Threshold' of 300 (red dashed line). In contrast, Kolkata has the lowest median in the group, but is showing a very large vertical spread (whiskers) which suggests high variability.

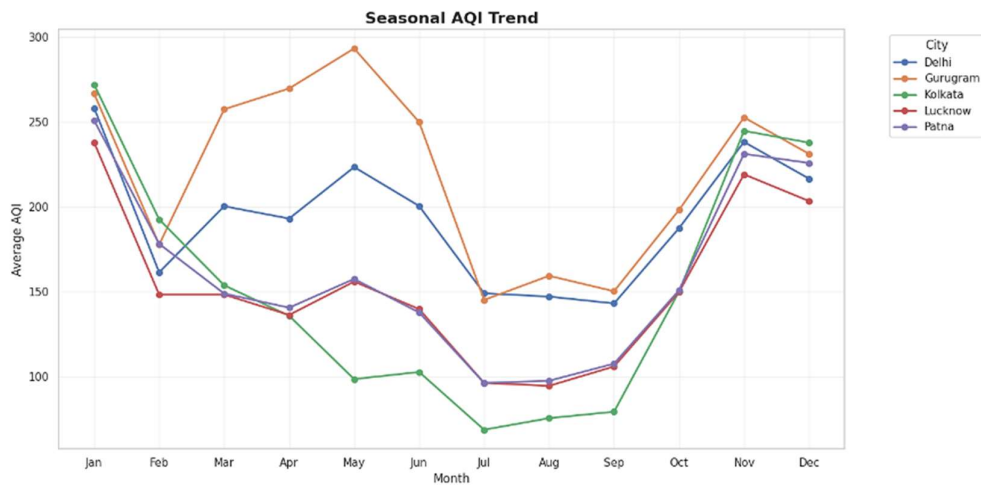


Figure 4.17: Comparative Monthly AQI Trends

- Seasonal AQI Trend Line Plot:
 - The chart shows that the universal trend is to drop drastically in all five cities during the monsoon months to a yearly minimum in July and August. The average AQI in Kolkata is the lowest during this time which is below 75, showing the effectiveness of the precipitation in "cleansing" the air of particulate matter.
 - In most of the cities, the air quality degrades moderately during the month of May, but in Gurugram, the air quality degrades massively in May every year with average AQI close to 290. This means that the city of Gurugram is one of the most vulnerable cities in the list with respect to pre-monsoon dust storms and high heat photochemical reactions from the summer to monsoon season.
 - All five cities experience a similar and dramatic increase in pollution beginning in October and reaching its peak in January. In this period, AQI of all the monitored cities remain consistently above 200 (Poor to Very Poor).

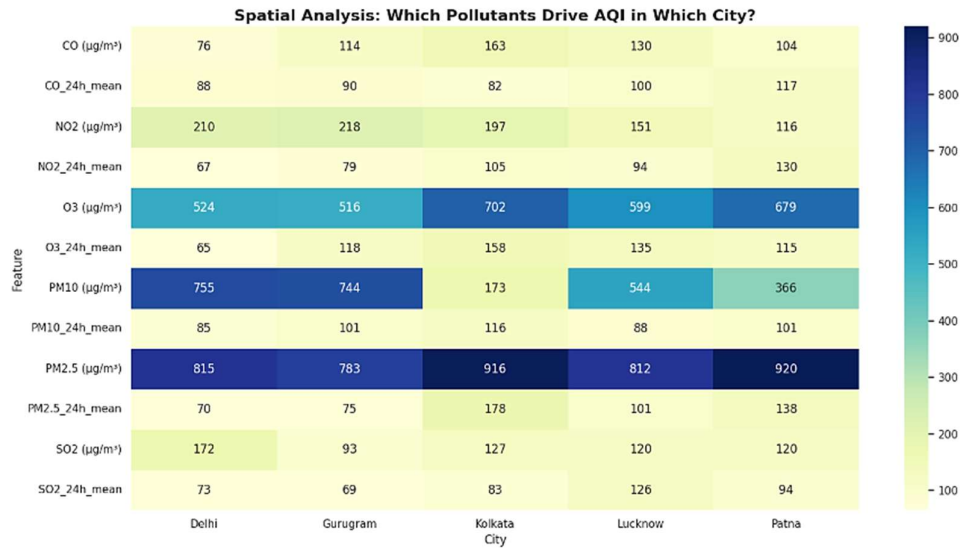


Figure 4.18: Feature Importance Heatmap by City

- Spatial Correlation:
 - PM2.5 is the most important contributor to AQI and has been found to have a value more than 800 (920 in Patna and 916 in Kolkata) across all hubs. The particulate matter features are given more importance in our predictive models, because in Indian urban environment, AQI changes are synonymous with PM2.5 concentration.
 - Globally, PM2.5 leads the way, and the secondary drivers are different across space. Delhi and Gurugram are highly sensitive to PM10 (755 and 744 respectively) possibly because of the construction dust and road re-suspension. In contrast, there is a much higher correlation of the meteorological conditions in Kolkata with O3 (Ozone) at 702, indicating that meteorological conditions, coastal versus in the NCR region, and the formation of secondary pollutants is of greater importance than in the NCR.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

The study was able to successfully build a very powerful localized air quality prediction model for the highly volatile Indian urban setting. The framework is able to cope with heavy, non-linear pollution increases in different cities due to a change from a global approach to a Weighted Ensemble approach that is spatially aware. The framework is effective when dealing with severe, non-linear peaks in pollution in different cities, because it is a spatial aware Weighted Ensemble approach instead of a generalized global model. The incorporation of engineered temporal features, rolling averages and specific chemical weightings were central to the stabilization of predictions from spurious momentary sensor noises and the capture of true diurnal cycles.

In addition, the use of SHAP analytics and interactive Power BI dashboards makes this complex machine learning model easily accessible and actionable. This system, which emphasizes explainability as well as accuracy, gives policy makers and public health executives a clear picture of the local chemical triggers that underlie each forecast.

5.2 Future Scope

- Future versions of this framework could incorporate real-time satellite aerosol optical depth (AOD) data, like that from the NASA's MODIS or Sentinel-5P. If ground level data and orbital observations were combined, the spatial resolution would be significantly improved and the model would be able to provide AQI for unmonitored rural areas between cities.
- The current Weighted Ensemble is very interpretable and stable, but further investigation of deep learning models such as Graph Neural Networks (GNNs) or Spatiotemporal LSTMs might be warranted. The architectures could mathematically simulate the actual 'flow' of pollutants blown by the wind over the Indo-Gangetic plain (such as the effect of agricultural burning in Punjab on Delhi days later).
- In addition to predicting the overall chemical concentration, the model can be expanded to predict the actual sources of the pollution: Real-Time Source Apportionment. The framework would combine live traffic API data with industrial thermal anomaly data to dynamically indicate to the policymaker if a sudden rise in PM_{2.5} was due to increased traffic in the area or to a smog from industry.

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